

Ryo Kitano

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EDUCATION

University of Waterloo

Bachelor of Mathematics, Honours, Co-operative Program

Waterloo, ON

Sep. 2024 – May 2029

- **Award:** President's Scholarship (2024)
- **Coursework:** Optimization, Probability, Statistics, Linear Algebra II, Combinatorics, Graph Theory, Network Flow Theory, Algorithm Design & Data Abstraction

EXPERIENCE

Co-Founder & Technical Lead

Mar. 2026 – Present

Neural Point Analytica (NPA)

Remote / Japan

- Co-founded a 3-person software company and lead engineering for an NDA-protected enterprise workflow platform; secured its first recurring client contract worth **US\$1.5K/month**
- Lead architecture and delivery of auditable RFQ intake, quotation, reconciliation, and reporting workflows using **Next.js, TypeScript, PostgreSQL, and AWS**; turn client feedback into staged releases with Docker and GitHub Actions

Java Backend & Computer Vision Engineer Intern

May 2026 – Sep. 2026

AISTGroup LLC

Remote / Baku, Azerbaijan

- Delivered **Spring Boot** REST APIs for invoice, receipt, and file workflows in a 5-engineer team, spanning JPA entities, validation, localization, **Liquibase** migrations, and JUnit tests
- Implemented encrypted SMTP configuration and a **transactional outbox** that decoupled receipt delivery from API requests and enabled retryable asynchronous processing; hardened behavior through code review and tests
- Owned data preparation, training, and evaluation for a YOLO26-s + EfficientNet-B0 pipeline using **41.9K images and 180K annotated signs**; achieved **0.871 mAP@50** detection and **89.0% Top-1** classification across 154 sign classes

PROJECTS

🔗 Hybrid Token-Efficient Routing Agent | *Python, Qwen, Fireworks, Docker*

Jul. 2026

- Trained a hashed n-gram logistic router on **360 measured task outcomes** to route between **quantized Qwen2.5-1.5B** and a hosted LLM under a 2-vCPU / 4-GB budget; added deterministic verification and failover
- Passed **80/80** mixed-task evaluations using 9,745 tokens; on a 20-case hard set, optimized escalation prompting raised accuracy from **85% to 95%** while cutting token use by **24.6%**

🔗 KenMemory | [Live Site](#) | *OpenAI API, FastAPI, React, PostgreSQL/pgvector*

Mar. 2026 – Aug. 2026

- Independently built and deployed a privacy-aware RAG product; embedded only consented memories in **PostgreSQL/pgvector**, used **HNSW vector search** for semantic retrieval, and streamed grounded answers, verified by **63 backend tests**

🔗 ThetaTrap | *Python, Qwen, Alpaca MCP, SQLite*

Aug. 2026 – Sep. 2026

- Independently built and deployed an MCP-native paper-options agent with **deterministic Python** controls that locked candidates, strikes, sizing, and max loss before LLM tool use; passed **191 tests** and **5/5 mutation-free replays**

🔗 Kraken Knight | *Python, Kraken API, SQLite, systemd*

Aug. 2026 – Sep. 2026

- Backtested a deterministic BTC/CAD strategy on **4.63M historical Kraken trades**, validated it through shadow trading, then deployed live execution with authenticated orders, fill reconciliation, idempotent audit logs, and fail-closed risk controls

TECHNICAL SKILLS

Languages: Python, Java, SQL; C, TypeScript, JavaScript (working knowledge)

AI/ML: Machine/deep learning, PyTorch, scikit-learn, Hugging Face Transformers, LLMs, RAG/vector search, prompt engineering, agentic/multi-agent systems, MCP/tool calling, computer vision, responsible AI, LLM evaluation

Frameworks & Data: TensorFlow/Keras, OpenAI API, FastAPI, React, Next.js, PostgreSQL/pgvector, Pinecone, pandas, NumPy, Matplotlib, Jupyter, REST APIs

MLOps & Developer Tools: Docker, GitHub Actions (CI/CD), AWS, Vercel, Render, Linux, Git/GitHub; Azure, GCP, Kubernetes (working knowledge)